

Total marks – 120**Attempt Question 1-8****All questions are of equal value**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 1 (15 marks)**Marks**

(a) Find

$$(i) \quad \int \frac{\cos \theta}{\sin^5 \theta} d\theta \quad 2$$

$$(ii) \quad \int \frac{dx}{x^2 + 2x + 2} \quad 2$$

$$(b) \quad \text{Use the substitution } t = \tan \frac{\theta}{2} \text{ to find } \int \frac{dx}{5 + 4\cos x + 3\sin x} \quad 3$$

$$(c) \quad \text{Use the substitution } u = -x \text{ to evaluate } \int_{-1}^1 \frac{dx}{e^x + 1} \quad 3$$

(d) Evaluate the following definite integrals:

$$(i) \quad \int_0^1 \cos^{-1} x \, dx \quad 2$$

$$(ii) \quad \int_1^2 x(\ln x)^2 \, dx \quad 3$$

Question 2 (15 marks) Start a new booklet

(a) If $z = 3 - 2i$ mark clearly on an Argand diagram the points represented by,

(i) $2z$ 1

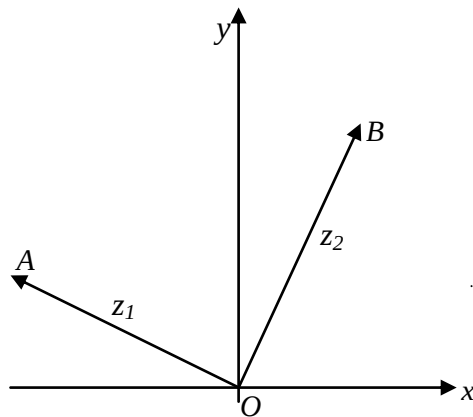
(ii) $-2iz$ 2

(b) z is a complex number such that $\arg z = \frac{\pi}{3}$ and $|z| \leq 2$.

(i) Sketch the locus of the point P representing z in the Argand diagram. 2

(ii) Find the possible values of the principal argument of $z - 1$ for z on this locus. 2

(c)



In the Argand diagram, vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent the complex numbers

$$z_1 = 2\left(\cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}\right) \text{ and } z_2 = 2\left(\cos \frac{7\pi}{15} + i \sin \frac{7\pi}{15}\right) \text{ respectively.}$$

(i) Show that $\triangle OAB$ is equilateral 3

(ii) Explain why $z_2 - z_1$ is equal to z_2 rotated by $\frac{\pi}{3}$ radians in a clockwise direction 2

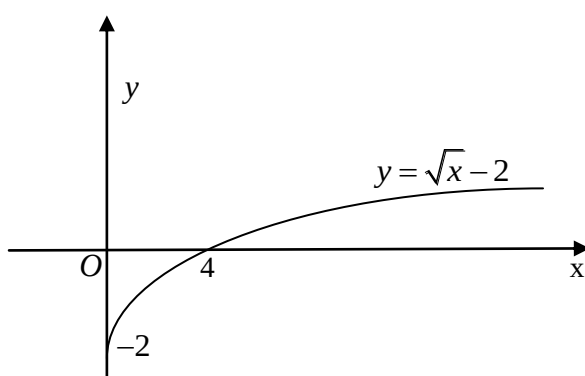
(iii) Express $z_2 - z_1$ in modulus-argument form. 3

Question 3 (15 marks) Start a new booklet

- (a) The polynomial $p(x) = x^4 - 2x^3 + 2x - 1$ has a root of multiplicity 3.
Find this root and hence factorise $p(x)$ **3**

- (b) Sketch the curve $y^2 = x^2(1 - x^2)$ clearly showing all relevant details. **6**

(c)

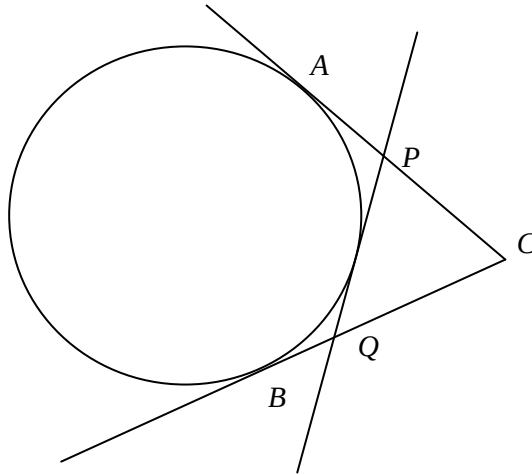


The diagram shows the graph of the function $f(x) = \sqrt{x} - 2$. On separate diagrams (each of half a page) sketch the following graphs, showing clearly any intercepts on the coordinate axes and the equations of any asymptotes:

- | | | |
|-------|----------------------|----------|
| (i) | $y = f(x) $ | 1 |
| (ii) | $y = [f(x)]^2$ | 1 |
| (iii) | $y = \frac{1}{f(x)}$ | 2 |
| (iv) | $y = \ln f(x)$ | 2 |

Question 4 (15 marks) Start a new booklet

- (a) A and B are two points on a circle. Tangents at A and B meet at C . A third tangent cuts CA and CB in P and Q respectively, as shown in the diagram. **3**
 Show that the perimeter of $\triangle CPQ$ is independent of PQ .



- (b) The polynomial $P(x)$ leaves a remainder of 9 when divided by $(x-2)$ and a remainder of 4 when divided by $(x-3)$. Find the remainder when $P(x)$ is divided by $(x-2)(x-3)$. **4**
- (c) The polynomial $P(x) = x^4 + 3x^3 + 6x^2 + 12x + 8$ has one root $2i$. **4**
 Find all the roots of $P(x)$.
- (d) If α, β are the roots of the equation $x^2 - px + q = 0$ and $S_n = \alpha^n + \beta^n$ where n is a positive integer, show that **4**

$$S_{n+2} - pS_{n+1} + qS_n = 0$$
 Hence, or otherwise find S_3, S_4 in terms of p, q .

Question 5 (15 marks) Start a new booklet

(a) $P\left(3p, \frac{3}{p}\right)$ and $Q\left(3q, \frac{3}{q}\right)$ are points on different branches of the hyperbola $xy = 9$.

(i) Find the equation of the tangent at P . 2

(ii) Find the point of intersection, T , of the tangents at P and Q . 2

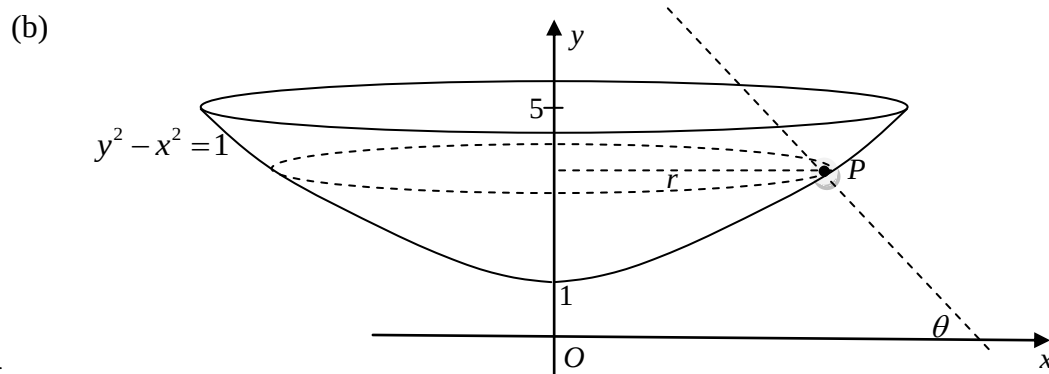
(iii) If the chord PQ passes through the point $(0, 2)$, find the locus of T . 3

(b) The region bounded by the graphs of $y = x^2$ and $y = x + 2$ is revolved 4
around the line $x = 3$. Derive the volume of the resulting solid as a definite
integral. **Do not calculate the value of this integral.**

(c) A solid has, as its base, the circular region in the xy -plane bounded by the 4
graph of $x^2 + y^2 = a^2$, where $a > 0$. If every cross-section by a plane
perpendicular to the x -axis is an equilateral triangle, with one side in the
base, show that the volume of the solid is $\frac{4\sqrt{3}}{3}a^3$ units³.

Question 6 (15 marks) Start a new booklet

- (a) A particle of mass m moves in a straight line away from a fixed point O in the line, such that at time t its displacement from O is x and its velocity is v . **4**
 At time $t = 0$, $x = 1$ and $v = 0$. Subsequently, the only force acting on the particle is one of magnitude $m\frac{k}{x^2}$, where k is a positive constant in a direction away from O . Show that v cannot exceed $\sqrt{(2k)}$.



A bowl is formed by rotating the hyperbola $y^2 - x^2 = 1$ for $1 \leq y \leq 5$ about the y axis. A particle P of mass m moves around the inner surface of the bowl in a horizontal circle with constant angular velocity ω .

- (i) Show that if the radius of the circle in which P moves is r , then the normal to the surface at P makes an angle θ with the horizontal as shown, where $\tan \theta = \frac{\sqrt{1+r^2}}{r}$. **4**
- (ii) Draw a diagram showing the forces acting on P . **1**
- (iii) By resolving these forces in the horizontal and vertical directions show that $r = \frac{\sqrt{g^2 - \omega^4}}{\omega^2}$ and the normal reaction $N = m\sqrt{2g^2 - \omega^4}$ **3**
- (iv) Since P must be in contact with the surface of the bowl and the radius must be positive, prove $\sqrt{\frac{g}{5}} \leq \omega \leq \sqrt{g}$. **3**

Question 7 (15 marks) Start a new booklet

a) i) Expand $(2+i)(3+i)$ 1

ii) By considering the arguments of each side, or otherwise show that 2
 $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4}$

iii) By first expanding $(p+q+i)(p^2+pq+1+iq)$ derive 4
 $\tan^{-1}\left(\frac{1}{p+q}\right) + \tan^{-1}\left(\frac{q}{p^2+pq+1}\right) = \tan^{-1}\left(\frac{1}{p}\right)$

b) i) Prove, by Mathematical Induction, that 3
 $\sin \theta = 2^n \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^n} \sin \frac{\theta}{2^n}$ for any positive integer n

ii) Hence show that $\frac{\sin \theta}{\theta} = \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \left(\frac{\theta}{2^n}\right) \frac{\sin \left(\frac{\theta}{2^n}\right)}{\left(\frac{\theta}{2^n}\right)}$ 1

iii) Hence find $\lim_{n \rightarrow \infty} \left(\frac{\sin \theta}{\theta}\right)$ 1

iv) Using $\prod$ as the product of terms, show that when $\theta = \frac{\pi}{2}$ 1

$$\prod_{k=2}^{\infty} \cos \left(\frac{\pi}{2^k}\right) = \frac{2}{\pi}$$

v) Hence show, by applying the half angle formula for $\cos \left(\frac{\theta}{2}\right)$ 2

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2+\sqrt{2}}}{2} \times \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \times \dots$$

Question 8 (15 marks) Start a new booklet

- (a) The ellipse $\mathcal{E}$: $\left(\frac{x}{5}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$ has foci $S(4, 0)$ and $S'(-4, 0)$.
- (i) Sketch the ellipse $\mathcal{E}$ indicating its foci S, S' and its directrices. 1
- (ii) Find the tangent at $P(x_1, y_1)$ on the ellipse $\mathcal{E}$. 1
- (iii) The line joining $P(x_1, y_1)$ to $Q(x_2, y_2)$ passes through S . Show that 2
- $$4(y_2 - y_1) = x_1 y_2 - x_2 y_1.$$
- (iv) If it is also known that $Q(x_2, y_2)$ lies on $\mathcal{E}$ find the point of 1
- intersection of the tangents at P and Q
- (v) By using the result in (iii) show the tangents at P and Q intersect on 2
- the directrix corresponding to S .
- (b) $I_n = \int_1^e (1 - \ln x)^n dx, \quad n = 1, 2, 3, \dots$
- (i) Show $I_n = -1 + nI_{n-1}, \quad n = 1, 2, 3, \dots$ 3
- (ii) Hence evaluate $\int_1^e (1 - \ln x)^3 dx$. 2
- (iii) Show that $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}, \quad n = 1, 2, 3, \dots$ 3

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a \neq 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, x > 0$

Solutions

Question 1

(a) (i) $u = \sin \theta, \quad du = \cos \theta d\theta$

$$\int \frac{du}{u^5} = -\frac{1}{4}u^{-4} + c$$

$$= -\frac{1}{4}\sin^{-4} \theta + c$$

(ii) $\int \frac{dx}{(x+1)^2 + 1} = \tan^{-1}(x+1) + c$

(b) $t = \tan \frac{x}{2}, \quad \therefore \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} = \frac{1}{2}(1+t^2)$

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\int \frac{\frac{2dt}{1+t^2}}{5 + \frac{4(1-t^2)}{1+t^2} + \frac{6t}{1+t^2}}$$

$$= \int \frac{2dt}{t^2 + 6t + 9} = \int \frac{2dt}{(t+3)^2}$$

$$= 2(t+3)^{-1} + c$$

$$= \frac{-2}{\tan \frac{x}{2} + 3} + c$$

(c) $u = -x, \quad -du = dx$

When $x = -1, u = 1$ and when $x = 1, u = -1$

$$\int_1^{-1} \frac{-du}{e^{-u} + 1} = \int_{-1}^1 \frac{du}{e^{-u} + 1} = \int_{-1}^1 \frac{du}{\frac{1}{e^u} + 1} = \int_{-1}^1 \frac{e^u du}{1 + e^u}$$

$$= \left[\ln(e^u + 1) \right]_{-1}^1 = \ln \frac{e+1}{1/e+1} = \ln e = 1$$

d)i) $\int_0^1 \cos^{-1} x \frac{d}{dx}(x) dx = x \cos^{-1} x \Big|_0^1 - \int_0^1 \frac{-x}{\sqrt{1-x^2}} dx$

$$= 0 - \frac{1}{2} \int_0^1 \frac{-2x}{\sqrt{1-x^2}} dx = \sqrt{1-x^2} \Big|_0^1 = 1$$

OR

$$\text{Area} = \int_0^{\pi/2} \sin y \, dy = [-\cos y]_0^{\pi/2} = 1$$

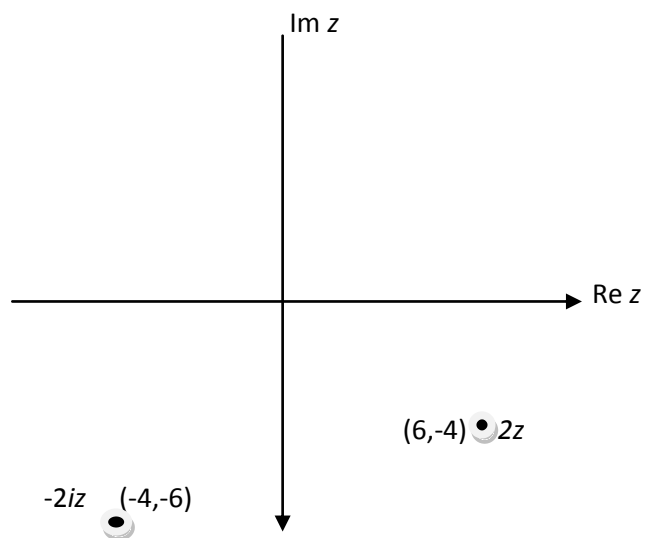
(ii) $\left[\frac{1}{2} x^2 (\ln x)^2 \right]_1^2 - \int_1^2 \frac{1}{2} x^2 \cdot 2 \cdot \frac{1}{x} \ln x \, dx$

$$= 2(\ln 2)^2 - \int_1^2 x \ln x \, dx$$

$$\begin{aligned}
 &= 2(\ln 2)^2 - \left[\frac{1}{2} x^2 \cdot \ln x \right]_1^2 + \int_1^2 x^2 \cdot \frac{1}{x} dx \\
 &= 2(\ln 2)^2 - 2 \ln 2 + \left[\frac{1}{4} x^2 \right]_1^2 \\
 &= 2(\ln 2)^2 - 2 \ln 2 + \frac{3}{4}
 \end{aligned}$$

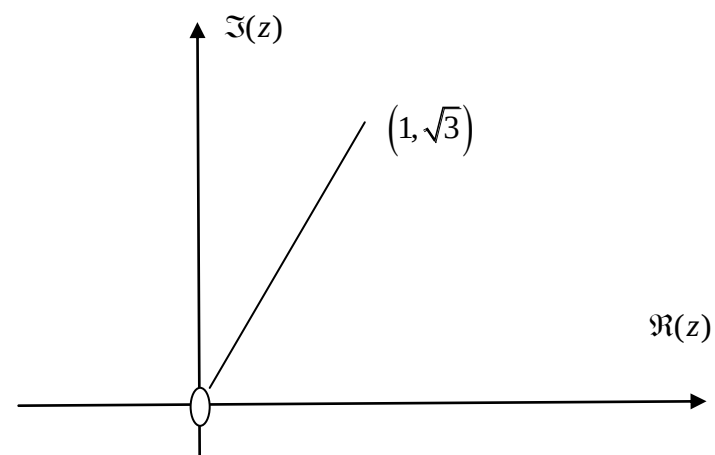
Question 2

(a)



(b)

i)



$$(ii) \quad \frac{\pi}{2} \leq \arg(z-1) < \pi$$

$$(i) \quad \therefore \arg \angle AOB = \arg z_1 - \arg z_2 = \frac{4\pi}{5} - \frac{7\pi}{15} = \frac{\pi}{3}$$

$$OA = OB = 2$$

Since Triangle is Isosceles and the included angle is $\pi/3$

$\therefore \triangle OAB$ is equilateral.

(ii) The vector $\overrightarrow{AB}$ represents $z_2 - z_1$. Now, $\overrightarrow{AB}$ is a clockwise rotation of $\overrightarrow{OB}$ by $\frac{\pi}{3}$ since $|AB| = |OB|$ and $\angle AOB = \frac{\pi}{3}$

iii)

$$\begin{aligned}\therefore z_2 - z_1 &= z_2 \left(\cos \frac{-\pi}{3} + i \sin \frac{-\pi}{3} \right) \\ &= 2 \left(\cos \frac{7\pi}{15} + i \sin \frac{7\pi}{15} \right) \left(\cos \left(-\frac{\pi}{3} \right) + i \sin \left(-\frac{\pi}{3} \right) \right) \\ &= 2 \left(\cos \frac{2\pi}{15} + i \sin \frac{2\pi}{15} \right)\end{aligned}$$

Question 3

(a) $p(x) = x^4 - 2x^3 + 2x - 1$
 $\therefore p'(x) = 4x^3 - 6x^2 + 2$
 $\therefore p''(x) = 12x^2 - 12x = 0$
when $x = 0, 1$
 $p'(1) = 4(1)^3 - 6(1)^2 + 2 = 0$
and $p(1) = 1 - 2 + 2 - 1 = 0$
 $\therefore x = 1$ is the triple root
 $\sum \alpha = +2 = 1 + 1 + 1 + \alpha$
 $\therefore$ the fourth root $\alpha = -1$
 $\therefore$ roots are 1, 1, 1, -1.

$$b) y^2 = x^2(1 - x^2)$$

$$\text{Since } y \geq 0 \quad x^2 \leq 1 \Rightarrow |x| \leq 1$$

$$\text{when } y = 0 \quad x = 0, \pm 1$$

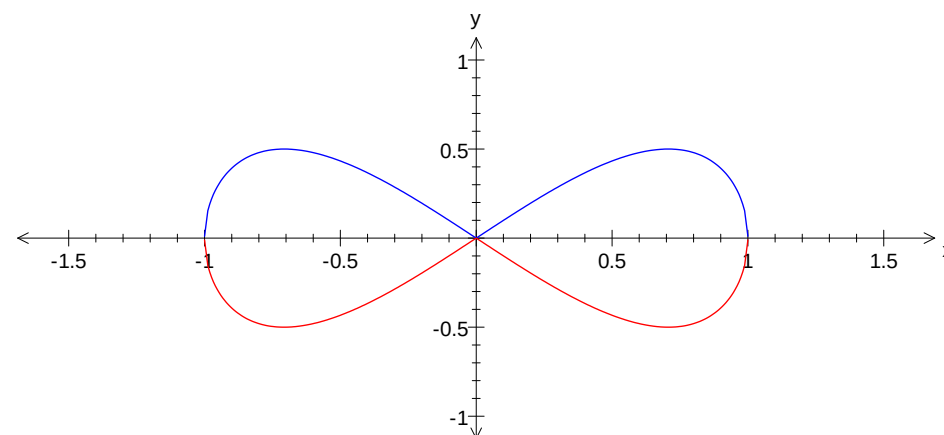
$$y \frac{dy}{dx} = 2x - 4x^3 = 2x(1 - 2x^2)$$

$$\therefore \frac{dy}{dx} = \frac{2x(1 - 2x^2)}{\pm \sqrt{x^2(1 - x^2)}} = \frac{2(1 - 2x^2)}{\pm \sqrt{1 - x^2}}$$

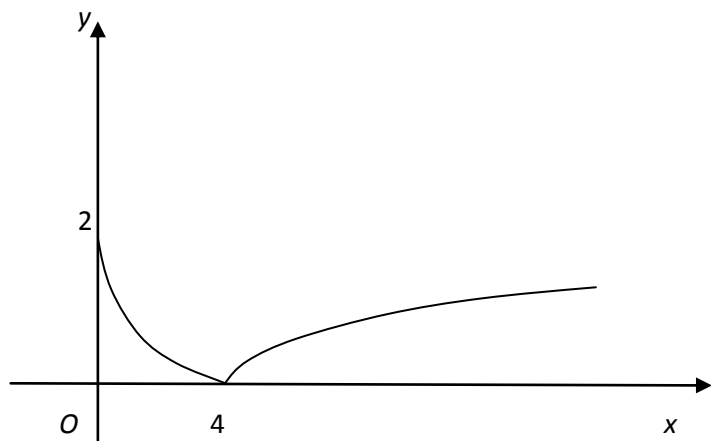
$$= 0 \text{ when } x = \pm \frac{1}{\sqrt{2}} \Rightarrow y = \pm \frac{1}{2}$$

$$\text{when } x \rightarrow \pm 1 \quad \frac{dy}{dx} \rightarrow \infty \therefore \text{vertical tangent}$$

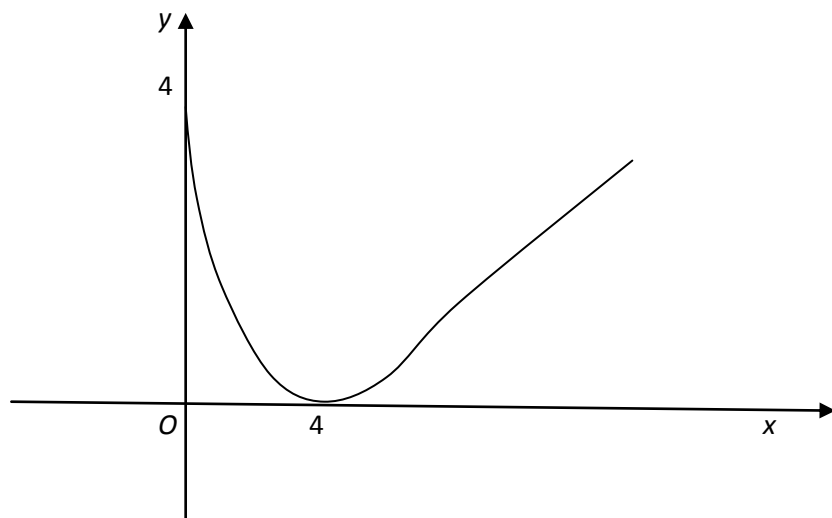
$$\text{at } x = 0 \quad \frac{dy}{dx} = \pm 1 \therefore \text{tangents inclined at } \pm 45^\circ$$



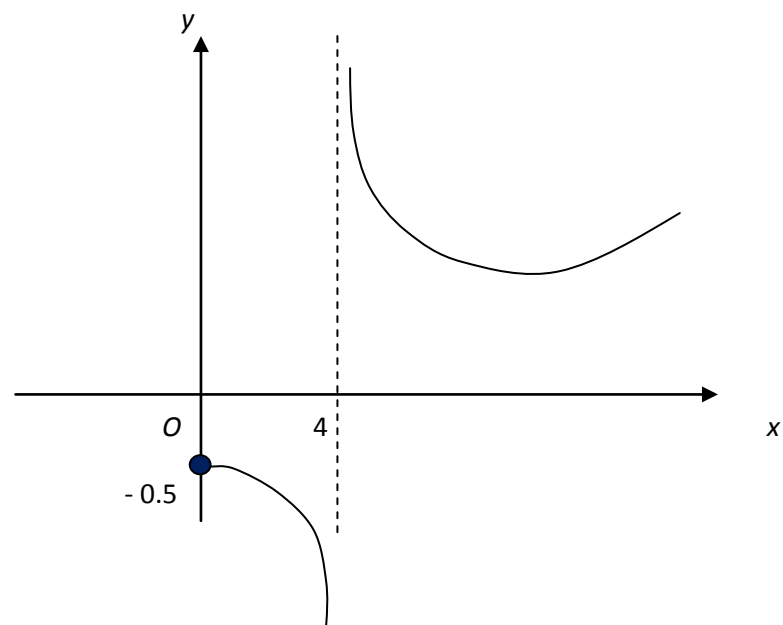
(c)(i) $y = |f(x)|$



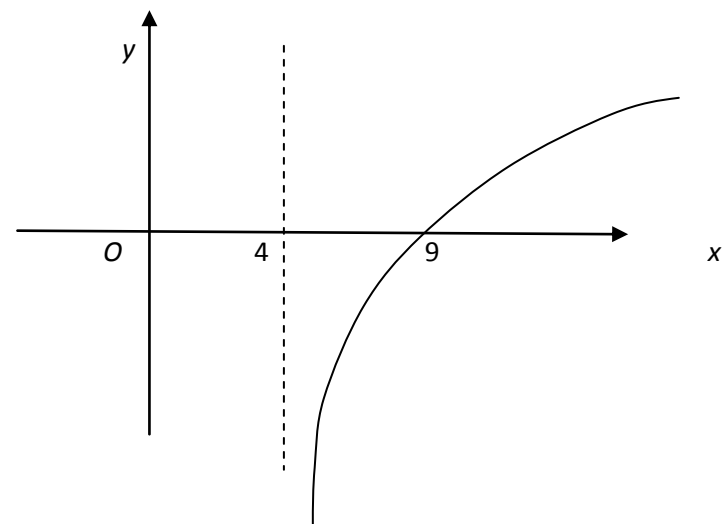
(ii) $y = [f(x)]^2$



(iii) $y = \frac{1}{f(x)}$



(iv) $y = \ln f(x)$



Question 4

(a) Perimeter of $\triangle CPQ = CP + CQ + PQ$

But $PQ = AP + BQ$ (tangents drawn from P are of equal length and
(tangents drawn from Q are of equal length)

$$\text{Perimeter of } \triangle CPQ = CP + AP + CQ + BQ = CA + CB$$

Which is independent of PQ .

$$b) \quad P(2) = 9, P(3) = 4$$

$$P(x) = Q(x) \cdot (x-2)(x-3) + R(x), \text{ where } R(x) = ax + b$$

$$P(2) = 0 + 2a + b = 9$$

$$P(3) = 0 + 3a + b = 4$$

$$a = -5, b = 19$$

Remainder is $19 - 5x$

c) Since all coefficients are real if $2i$ is a root, so is the conjugate $-2i$

$$\text{Let other 2 roots be } \alpha, \beta \therefore 2i + -2i + \alpha + \beta = \alpha + \beta = -3$$

$$\prod \alpha = 4\alpha\beta = 8$$

$$\text{solve simultaneously } 4\alpha(-3-\alpha) = -12\alpha - 4\alpha^2 = 8$$

$$\Rightarrow \alpha^2 + 3\alpha + 2 = 0 \Rightarrow \alpha = -1, -2$$

$$\therefore P(x) = (x-2i)(x+2i)(x+1)(x+2)$$

d)

$$d) \quad x^2 - px + q = 0, \text{ if roots are } \alpha, \beta$$

$$\alpha^2 - p\alpha + q = 0 \text{ and } \beta^2 - p\beta + q = 0 \Rightarrow \alpha^2 + \beta^2 = p(\alpha + \beta) - 2q = p^2 - 2q = S_2$$

$$\text{also } \alpha^n(\alpha^2 - p\alpha + q) = 0 \quad \beta^n(\beta^2 - p\beta + q) = 0$$

$$\text{i.e. } \alpha^{n+2} - p\alpha^{n+1} + q\alpha^n = 0 \quad \beta^{n+2} - p\beta^{n+1} + q\beta^n = 0 \text{ add these results}$$

$$S_{n+2} - pS_{n+1} + qS_n = 0$$

$$S_3 - pS_2 + qS_1 = S_3 - p(p^2 - 2q) + q(p) = 0 \Rightarrow S_3 = p^3 - 3pq$$

$$S_4 - pS_3 + qS_2 = S_4 - p(p^3 - 3pq) + q(p^2 - 2q) = 0$$

$$S_4 = p^4 - 4p^2q + 2q^2$$

Question 5

$$(a) \quad (i) \quad x \frac{dy}{dx} + y = 0, \therefore \frac{dy}{dx} = \frac{-y}{x}$$

$$\text{At } P, \frac{dy}{dx} = \frac{-3/p}{3p} = -\frac{1}{p^2}$$

$$\text{Required equation: } y - \frac{3}{p} = -\frac{1}{p}(x - 3p)$$

$$\text{Which gives } x + p^2y = 6p$$

$$(ii) \quad \text{tangent at } P \quad x + p^2y = 6p$$

$$\text{Tangent at } Q \quad x + q^2y = 6q$$

When solved simultaneously, we get the coordinates of T:

$$\left(\frac{6pq}{p+q}, \frac{6}{p+q} \right)$$

$$(iii) \quad m_{PQ} = \frac{\frac{3}{p} - \frac{3}{q}}{3p - 3q} = -\frac{1}{pq}$$

$$\text{Equation } PQ: \quad y - \frac{3}{p} = -\frac{1}{pq}(x - 3p)$$

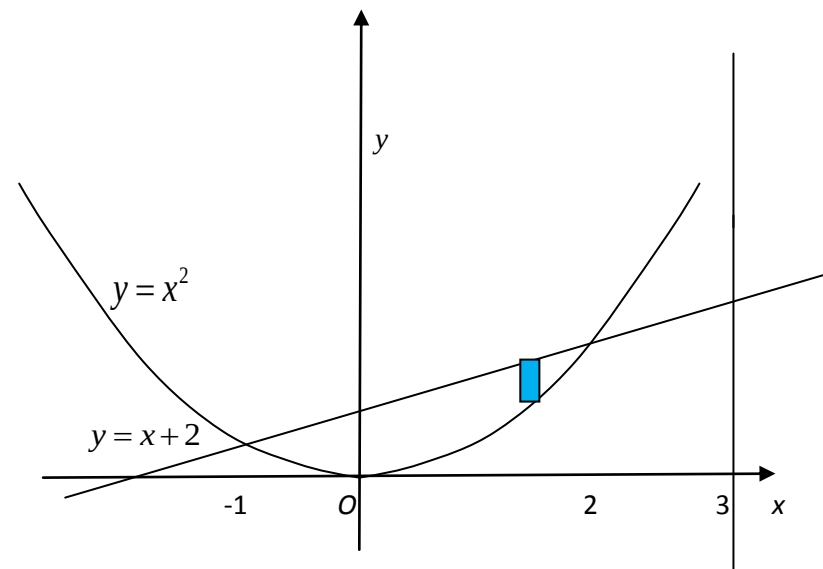
Now, when $x = 0, y = 2$

$$\therefore \frac{p+q}{pq} = \frac{2}{3} \text{ or } p+q = \frac{2pq}{3}$$

$$\text{At } T, \quad x = \frac{6pq}{p+q} = \frac{6pq}{\frac{2pq}{3}} = 9$$

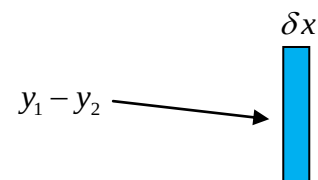
Therefore the locus is $x = 9$

b)



To find points of intersection: $x^2 = x + 2, \therefore x = -1, 2$

Consider a typical strip



Rotate the strip to form a shell. The volume of the shell is given by

$$\delta V = 2\pi rh \cdot \delta x, \text{ where } r = 3 - x \text{ and } h = y_1 - y_2 = x + 2 - x^2$$

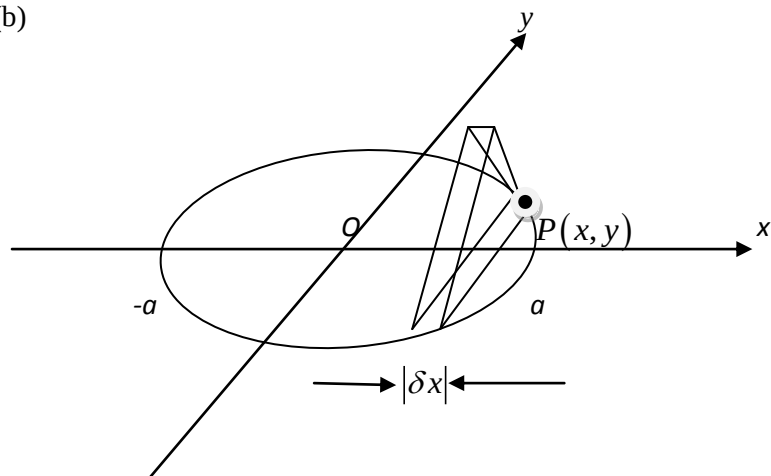
$$V \approx \sum_{x=-1}^{x=2} 2\pi(3-x)(x+2-x^2)\delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-1}^{x=2} 2\pi(3-x)(x+2-x^2)\delta x$$

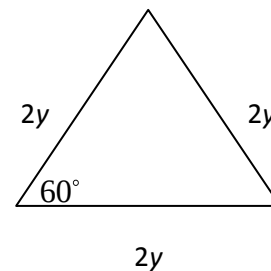
$$= \int_{-1}^2 2\pi(3-x)(x+2-x^2)dx$$

$$\text{OR} = \int_{-1}^2 \pi(y^2 - 11y + 6\sqrt{y} + 16)dy$$

(b)



Consider a typical slice of width δx .



$$\delta V = \frac{1}{2} 2y \cdot 2y \cdot \sin 60^\circ \cdot \delta x = y^2 \sqrt{3} \cdot \delta x$$

$$V \approx \sum_{x=-a}^{x=a} y^2 \sqrt{3} \cdot \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-a}^{x=a} y^2 \sqrt{3} \cdot \delta x = \lim_{\delta x \rightarrow 0} \sum_{x=-a}^{x=a} (a^2 - x^2) \sqrt{3} \cdot \delta x$$

$$= \sqrt{3} \int_{-a}^a (a^2 - x^2) dx$$

$$= 2\sqrt{3} \int_0^a (a^2 - x^2) dx$$

$$= 2\sqrt{3} \left[a^2 x - \frac{x^3}{3} \right]_0^a$$

$$= \frac{4\sqrt{3}}{3} a^3$$

Question 6

(a) Choose the initial direction as positive

$$\ddot{x} = \frac{k}{x^2}, \quad k > 0$$

$$v \frac{dv}{dx} = \frac{k}{x^2} \Rightarrow v dv = \frac{k}{x^2} dx$$

$$\frac{1}{2} v^2 = -\frac{k}{x} + c, \text{ where } c \text{ is constant}$$

$$\text{Now, when } x=1, v=0 \Rightarrow c=k$$

$$\therefore v^2 = 2k \left(1 - \frac{1}{x} \right)$$

$$\text{Now, } x \geq 1 \quad \therefore 0 \leq 1 - \frac{1}{x} < 1$$

$$\therefore 0 \leq v^2 < 2k$$

$$\text{Hence, } v \text{ cannot exceed } \sqrt{2k}$$

$$(b) (i) \quad y^2 - x^2 = 1 \Rightarrow 2y \frac{dy}{dx} - 2x = 0 \Rightarrow \frac{dy}{dx} = \frac{x}{y}$$

$$\text{At } P, \frac{dy}{dx} = \frac{r}{\sqrt{1+r^2}}$$

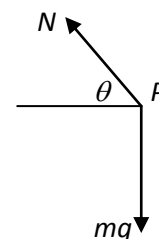
$$\text{Hence, the gradient of the normal at } P \text{ is } \frac{-\sqrt{1+r^2}}{r}$$

Now, the gradient of the normal is the tangent of the angle made with the 'positive' x axis.

$$\therefore \tan(180^\circ - \theta) = \frac{-\sqrt{1+r^2}}{r}$$

$$\therefore \tan \theta = \frac{\sqrt{1+r^2}}{r}$$

(ii)



(iii) Resolve forces

Horizontally

Vertically

$$mr\omega^2 = N \cos \theta$$

$$mg = N \sin \theta$$

$$\therefore \tan \theta = \frac{g}{r\omega^2} = \frac{\sqrt{1+r^2}}{r} \text{ from part (i)}$$

$$1+r^2 = \frac{g^2}{\omega^4} \Rightarrow r = \frac{\sqrt{g^2 - \omega^4}}{\omega^2}$$

$$\text{Now, } \cos^2 \theta + \sin^2 \theta = 1$$

$$\therefore \left(\frac{mr\omega^2}{N} \right)^2 + \left(\frac{mg}{N} \right)^2 = 1$$

Question 7

$$N^2 = m^2 r^2 \omega^4 + m^2 g^2 = m^2 \frac{g^2 - \omega^4}{\omega^4} \omega^4 + m^2 g^2$$

$$= m^2 (2g^2 - \omega^4)$$

$$\therefore N = m\sqrt{(2g^2 - \omega^4)}$$

$$(iv) \quad r = \frac{\sqrt{g^2 - \omega^4}}{\omega^2} \text{ and } r > 0$$

$$\therefore g^2 > \omega^4$$

$$\text{But } N > 0 \quad \therefore 2g^2 > \omega^4$$

Both these conditions exist if $g > \omega^2$

Note: $y \leq 5 \Rightarrow y^2 \leq 25 \Rightarrow 1 + r^2 \leq 25$

$$\therefore \frac{g^2}{\omega^4} > 25 \Rightarrow \omega \geq \sqrt{\frac{g}{5}}$$

$$\therefore \sqrt{\frac{g}{5}} \leq \omega \leq \sqrt{g}$$

$$a) i) (2+i)(3+i) = 6 - 1 + 2i + 3i = 5 + 5i$$

$$ii) \therefore \arg(2+i)(3+i) = \arg(5+5i)$$

$$\therefore \arg(2+i) + \arg(3+i) = \tan^{-1} 1$$

$$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{4}$$

$$a) iii) (p+q+i)(p^2 + pq + 1 + qi)$$

$$= p^3 + p^2q + p + pqi + p^2q + pq^2 + q + q^2i + p^2i + pqi + i - q$$

$$= p(p^2 + 2pq + q^2 + 1) + i(p^2 + 2pq + q^2 + 1)$$

$$\therefore \arg(p+q+i)(p^2 + pq + 1 + qi) = \arg(p(p^2 + 2pq + q^2 + 1) + i(p^2 + 2pq + q^2 + 1))$$

$$\therefore \tan^{-1} \frac{1}{p+q} + \tan^{-1} \frac{q}{p^2 + pq + 1} = \tan^{-1} \frac{1}{p} \text{ as required}$$

$$b)i) \text{ let } n=1 \text{ LHS} = \sin \theta \text{ RHS} = 2 \cos \frac{\theta}{2} \sin \frac{\theta}{2} = \sin \theta \therefore \text{ true when } n=1.$$

$$\text{assume true when } n=k \text{ i.e. } \sin \theta = 2^k \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \sin \frac{\theta}{2^k}$$

$$\text{now } \sin \frac{\theta}{2^k} = 2 \cos \frac{\theta}{2^{k+1}} \sin \frac{\theta}{2^{k+1}} \text{ substitute in assumption}$$

$$\text{thus } \sin \theta = 2^k \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \times 2 \cos \frac{\theta}{2^{k+1}} \sin \frac{\theta}{2^{k+1}}$$

$$\Rightarrow \sin \theta = 2^{k+1} \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \cos \frac{\theta}{2^{k+1}} \sin \frac{\theta}{2^{k+1}}$$

$\therefore$ if the result is true for $n=k$ it is also true for $n=k+1$

but it is true for $n=1 \therefore$ it is true for $n=1+1=2$ and so on for all positive integer n

b)ii) Divide both sides by θ

$$\text{thus } \frac{\sin \theta}{\theta} = \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^n} \sin \left(\frac{\theta}{2^n} \right)$$

$$\text{iii) as } n \rightarrow \infty \quad \frac{\theta}{2^n} \rightarrow 0 \quad \therefore \lim_{n \rightarrow \infty} \sin \left(\frac{\theta}{2^n} \right) = 1$$

$$\therefore \lim_{n \rightarrow \infty} \frac{\sin \theta}{\theta} = \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^n} \times 1$$

b)iii) Let $\theta = \frac{\pi}{2}$

$$\therefore \lim_{n \rightarrow \infty} \frac{\sin \theta}{\theta} = \frac{1}{\pi/2} = \cos \frac{\pi/2}{2} \cos \frac{\pi/2}{4} \dots \cos \frac{\pi/2}{2^n} = \prod_{k=2}^{\infty} \cos \frac{\pi}{2^k} = \frac{2}{\pi}$$

$$\text{b)iv) } \therefore \frac{2}{\pi} = \cos \frac{\pi}{4} \cos \frac{\pi}{8} \cos \frac{\pi}{16} \dots$$

$$\text{Now } \cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta) \Rightarrow \cos \theta = \frac{1}{\sqrt{2}} \sqrt{1 + \cos 2\theta}$$

$$\text{Now } \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

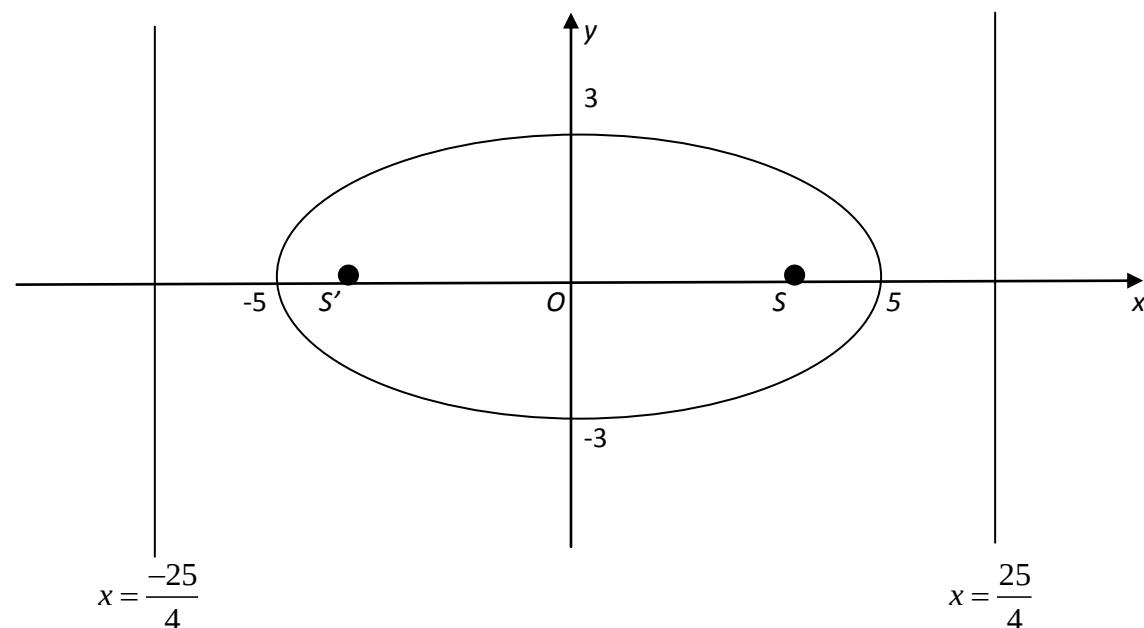
$$\therefore \cos \frac{\pi}{8} = \frac{1}{\sqrt{2}} \sqrt{1 + \cos \frac{\pi}{4}} = \frac{1}{\sqrt{2}} \sqrt{1 + \frac{\sqrt{2}}{2}} = \frac{1}{\sqrt{2}} \sqrt{\frac{2 + \sqrt{2}}{2}} = \frac{\sqrt{2 + \sqrt{2}}}{2}$$

$$\text{also } \cos \frac{\pi}{16} = \frac{1}{\sqrt{2}} \sqrt{1 + \cos \frac{\pi}{8}} = \frac{1}{\sqrt{2}} \sqrt{1 + \frac{\sqrt{2 + \sqrt{2}}}{2}} = \frac{\sqrt{2 + \sqrt{2 + \sqrt{2}}}}{2}$$

$$\therefore \frac{2}{\pi} = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2 + \sqrt{2}}}{2} \times \frac{\sqrt{2 + \sqrt{2 + \sqrt{2}}}}{2} \times \dots$$

Question 8

(a) (i)



$$ae = \pm 4, a = 5, \therefore e = \frac{4}{5}$$

Hence, the directrices are $x = \frac{\pm 25}{4}$

$$(ii) \quad \frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$\frac{2x}{25} + \frac{2y}{9} \frac{dy}{dx} = 1$$

At $P(x_1, y_1)$, $\frac{dy}{dx} = \frac{-9x_1}{25y_1}$

Required equation:

$$y - y_1 = \frac{-9x_1}{25y_1}(x - x_1)$$

$$9xx_1 + 25yy_1 = 9x_1^2 + 25y_1^2 = 225$$

Note: $P(x_1, y_1)$ lies on the curve $9x_1^2 + 25y_1^2 = 225$

$$(iii) \quad y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

At $x = 4, y = 0$

$$-y_1(x_2 - x_1) = (y_2 - y_1)(4 - x_1)$$

$$4(y_2 - y_1) = x_1(y_2 - y_1) - y_1(x_2 - x_1) = x_1y_2 - x_2y_1$$

(iv) Two equations are: $9xx_1 + 25yy_1 = 225$ and

$$9xx_2 + 25yy_2 = 225$$

$$25y = \frac{9xx_1 - 225}{y_1} = \frac{9xx_2 - 225}{y_2} \quad \therefore 9x(x_1y_2 - x_2y_1) = 225(y_2 - y_1)$$

$$\therefore 9x.4(y_2 - y_1) = 225(y_2 - y_1) \quad x = \frac{225}{36} = \frac{25}{4}$$

$$(i) \quad I_n = \int_1^e (1 - \ln x)^n dx$$

$$= \left[x(1 - \ln x)^n \right]_1^e - \int_1^e nx(1 - \ln x)^{n-1} \left(-\frac{1}{x} \right) dx$$

$$= -1 + nI_{n-1}$$

$$(ii) \quad I_3 = -1 + 3I_2 = -1 + 3(-1 + 2I_1) = -4 + 6(-1 + I_0)$$

$$= -10 + 6 \int_1^e dx = -10 + 6(e - 1) = -16 + 6e$$

$$(iii) \quad I_r = -1 + rI_{r-1}$$

$$\frac{I_r}{r!} = \frac{-1}{r!} + \frac{rI_{r-1}}{r!}$$

$$\frac{I_r}{r!} = \frac{-1}{r!} + \frac{I_{r-1}}{(r-1)!}$$

$$\sum_{r=1}^n \frac{I_r}{r!} = \sum_{r=1}^n \frac{-1}{r!} + \sum_{r=1}^n \frac{I_{r-1}}{(r-1)!}$$

$$\frac{I_n}{n!} + \sum_{r=1}^{n-1} \frac{I_r}{r!} = \sum_{r=1}^n \frac{-1}{r!} + \sum_{r=0}^{n-1} \frac{I_r}{r!}$$

$$\frac{I_n}{n!} = \sum_{r=1}^n \frac{-1}{r!} + \frac{I_0}{0!} = \sum_{r=1}^n \frac{-1}{r!} + \frac{\int_1^e dx}{0!} = \sum_{r=1}^n \frac{-1}{r!} + \frac{e}{0!} + \frac{-1}{0!} = e - \sum_{r=0}^n \frac{1}{r!}$$